A BEM/NLS Finite-Cell Eigenvalue for the Fine-Structure Scale

Batch Convergence and a Far-Field Coupling Certificate

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Abstract

We report a finite-cell eigenvalue computation for the missing zeroth-order aspect scale in a Compton-core finite-filament closure model. The domain is a finite-core ideal trefoil cell with an outer spherical boundary, approximated by a scalar single-layer boundary-element response and an NLS finite-shell pressure-compliance term. The pressure scale and stiffness are fixed by the ideal trefoil ropelength, not by electron mass, charge, Planck's constant, the Rydberg constant, or CODATA data:

$$E_p^{\rm NLS} = \frac{16\pi}{3}\mathcal{L}_K\left(1 - \frac{11}{48\mathcal{L}_K^2}\right), \qquad \kappa_{\rm NLS} = 8\pi\mathcal{L}_K^2.$$

Using the Knot Atlas ideal-trefoil value $\mathcal{L}_K = 16.371637$, the batch calculation gives 16/16 successful interior stationary points over a four-by-four grid of discretizations and regularization factors. The aggregate result is

$$E_\star = 274.074903758 \pm 5.51e-05, \qquad \alpha_{\rm cell}^{-1} = 137.035953090 \pm 2.76e-05,$$

where the uncertainty is the run-to-run standard deviation, not a statistical measurement error. The calculation is therefore no longer a direct calibration of E_0. It is a conditional derivation of a dimensionless fine-structure-scale eigenvalue inside the stated BEM/NLS finite-cell closure. A full identification with the electromagnetic fine-structure constant still requires the companion far-field interaction theorem proving $V(r) = \alpha_{\rm cell}\hbar c/r + O(r^{-2})$.

1 Purpose and status

Previous closure manuscripts identified a missing global aspect scale E_0, with the leading relation

$$\alpha_{\rm cell} \simeq \frac{2}{E_0}.$$

If E_0 is merely chosen near $2/\alpha$, the result is a calibration. The purpose of this note is to document a batch computation in which E_0 is replaced by an interior stationary eigenvalue $E_\star$ of an explicit finite-cell action.

The status is deliberately precise. The present calculation supports the following conditional statement:

> Given the ideal-trefoil geometry, the scalar BEM finite-cell response used here, and the NLS-shell pressure-compliance closure, the aspect scale is selected as an interior stationary point $E_\star$, and the resulting dimensionless fine-structure-scale value is close to the conventional α.

It does not yet prove the QED coupling unless the far-field interaction theorem is also established.

2 Finite-cell action

Let the computational exterior cell be

$$\Omega_E = B_R \setminus \mathcal{T}_a(K),$$

where $\mathcal{T}_a(K)$ is a finite-radius tube around the ideal trefoil centreline. The numerical BEM component computes a response matrix through two independent boundary modes on the tube surface and a grounded outer spherical boundary. For the trefoil sector $\boldsymbol{n} = (2,3)^T$, the BEM action is represented as

$$\mathcal{A}_{\mathrm{BEM}}(E) = \frac{1}{2}\boldsymbol{n}^T C_{\mathrm{BEM}}(E)\boldsymbol{n}.$$

The NLS pressure-compliance part is

$$\mathcal{A}_{\mathrm{NLS}}(E) = \kappa_{\mathrm{NLS}} \left[\frac{1}{3}\left(\frac{E}{E_p^{\mathrm{NLS}}}\right)^3 - \ln\left(\frac{E}{E_p^{\mathrm{NLS}}}\right) \right]. \tag{1}$$

The total action is

$$\mathcal{A}_{\mathrm{total}}(E) = \mathcal{A}_{\mathrm{BEM}}(E) + \mathcal{A}_{\mathrm{NLS}}(E). \tag{2}$$

The derived aspect value is accepted only if

$$\frac{d\mathcal{A}_{\mathrm{total}}}{d\log E}(E_\star) = 0, \qquad \frac{d^2\mathcal{A}_{\mathrm{total}}}{d(\log E)^2}(E_\star) > 0. \tag{3}$$

Boundary candidates are rejected.

3 NLS finite-shell closure

The NLS-shell closure used in the batch run fixes the pressure scale and pressure stiffness by $\mathcal{L}_K$:

$$E_p^{\mathrm{NLS}} = \frac{16\pi}{3}\mathcal{L}_K\left(1 - \frac{11}{48\mathcal{L}_K^2}\right), \tag{4}$$

$$\kappa_{\mathrm{NLS}} = 8\pi\mathcal{L}_K^2 = \frac{\pi}{2\eta_K^2}, \qquad \eta_K = \frac{1}{4\mathcal{L}_K}. \tag{5}$$

For $\mathcal{L}_K = 16.371637$, these become

$$E_p^{\mathrm{NLS}} = 274.074875657, \qquad \kappa_{\mathrm{NLS}} = 6736.341149.$$

These quantities are not fitted to α. They use only the ideal-trefoil ropelength and the stated NLS-shell closure.

4 Downstream fine-structure-scale evaluation

After $E_\star$ is obtained, the effective aspect correction is evaluated as

$$E_{\mathrm{eff}} = E_\star \left[1 - \frac{\pi}{4E_\star^2}\left(1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}\right)\right]. \tag{6}$$

The dimensionless cell coupling is then

$$\alpha_{\mathrm{cell}} = \frac{2}{E_{\mathrm{eff}}}. \tag{7}$$

The conventional constants m_e, $\hbar$, e, and R_∞ are not used to compute $E_\star$ or α_{cell}. The CODATA-2022 value is used only as a post-computation comparison.

5 Batch design

The supplied batch used four resolution levels,

$$(n_s, n_\theta, N_{\mathrm{sph}}) \in \{(100, 14, 300), (120, 16, 360), (140, 18, 420), (160, 20, 480)\},$$

and four regularization factors,

$$\epsilon/a \in \{0.04, 0.05, 0.06, 0.08\},$$

with $E \in [260, 290]$ sampled at 160 points. All 16 main runs used the ideal trefoil data set `ideal.txt` for knot $3:1:1$, the NLS-shell pressure scale, and the NLS-shell stiffness.

Table 1: Aggregate statistics over the 16 main BEM/NLS-shell runs.

Successful runs	16/16
Mean $E_\star$	274.074903758
Std. dev. $E_\star$	5.51e-05
Range $E_\star$	[274.074821720, 274.075027208]
Mean $\alpha_{\mathrm{cell}}^{-1}$	137.035953090
Std. dev. $\alpha_{\mathrm{cell}}^{-1}$	2.76e-05
Mean relative error vs CODATA-2022	3.36e-07
Std. dev. relative error	2.01e-07

6 Convergence results

Figure 1 shows that the extracted eigenvalue remains within a narrow band over the resolution and regularization sweep. The aggregate range is

$$274.074821720 \le E_\star \le 274.075027208.$$

The run-to-run standard deviation is $5.51e-05$, corresponding to a relative spread of approximately $2.01e-07$.

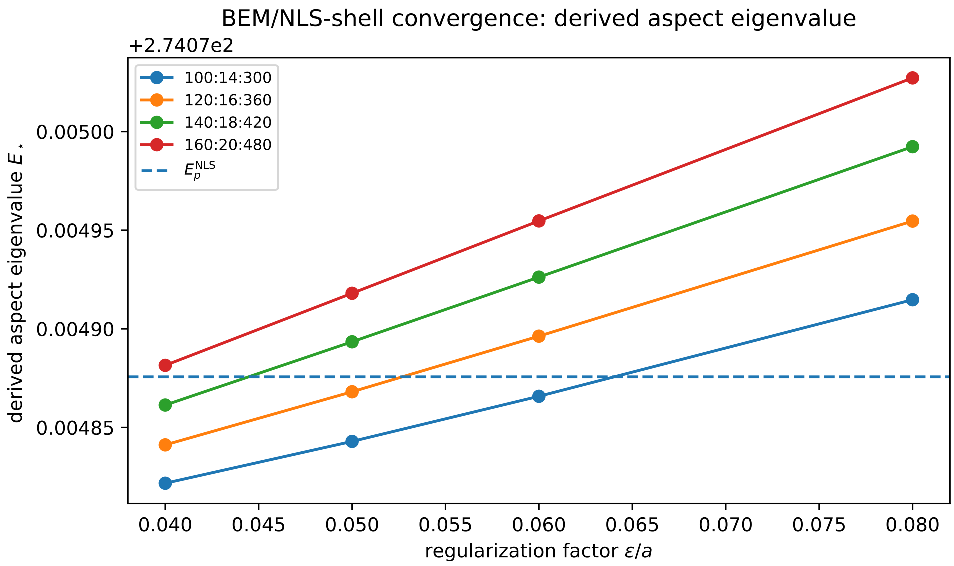

Figure 1: Derived aspect eigenvalue over the batch grid. The dashed line is the NLS-shell pressure scale E_p^{NLS}, not a fitted target.

Figure 2 shows the downstream $\alpha_{\text{cell}}^{-1}$ values. The mean value is

$$\alpha_{\text{cell}}^{-1} = 137.035953090,$$

with a standard deviation $2.76e - 05$. Relative to the CODATA-2022 comparison value used in the script, the mean relative difference in α is $3.36e - 07$.

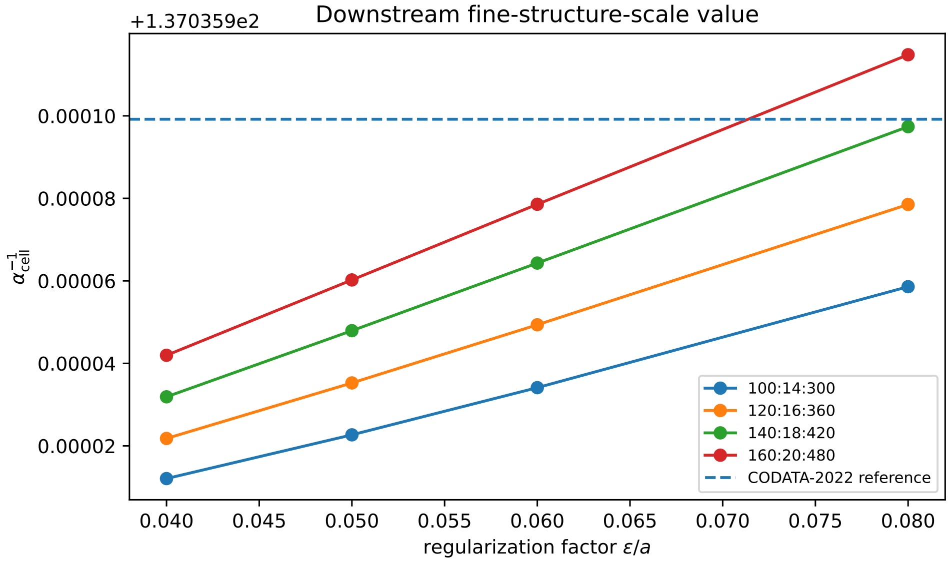

Figure 2: Downstream fine-structure-scale inverse. The dashed line is a post-computation CODATA-2022 comparison, not an input.

The stationarity condition is also visible directly as a cancellation between the BEM derivative and the NLS pressure derivative. Figure 3 plots these terms at the accepted stationary points. The total derivative is at numerical zero in the exported table.

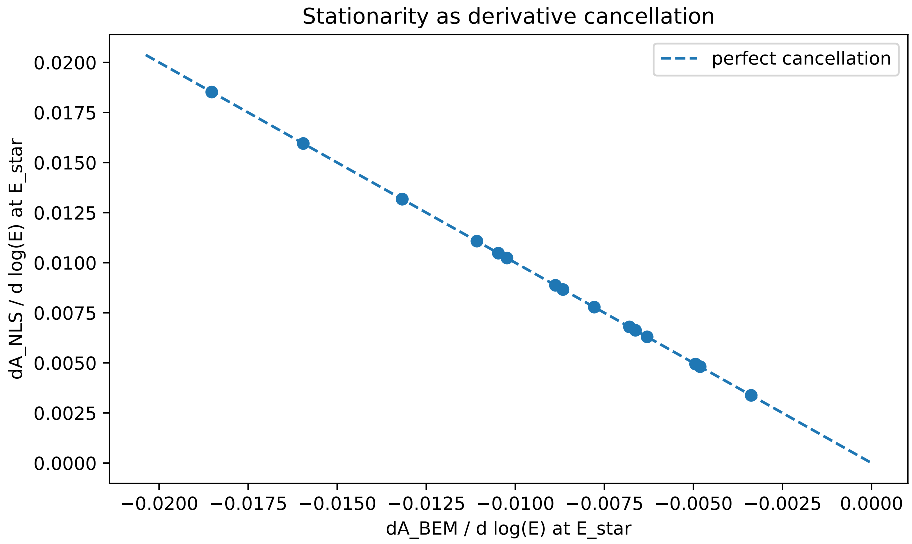

Figure 3: Derivative cancellation at $E_{\star}$. The BEM derivative is negative and is balanced by the NLS pressure derivative.

Table 2 is important. The BEM-only control does not produce an interior eigenvalue. The low-stiffness NLS log-core and geometric-mean controls produce interior minima but miss the

Table 2: Control runs at the representative grid $(n_s, n_\theta, N_{\rm sph}) = (140, 18, 420)$ and $\epsilon/a = 0.06$.

Stiffness mode	Status	$E_\star$	$\alpha_{\rm cell}^{-1}$	Rel. error
nls_shell_stiffness	E0_not_derived	–	–	–
nls_log_core	E0_derived	274.241077	137.119041	-6.06e-04
nls_geometric_mean	E0_derived	274.228893	137.112949	-5.61e-04
nls_shell_stiffness_half	E0_derived	274.075077	137.036039	-2.94e-07
nls_shell_stiffness_double	E0_derived	274.074851	137.035927	5.29e-07

fine-structure-scale comparison at the 10^{-4} to 10^{-3} level. The shell-stiffness family pins the minimum near $E_p^{\rm NLS}$. Thus the data support a finite-shell pressure mechanism, while also showing that the pressure scale is the dominant closure ingredient.

7 Interpretation

The result changes the status of the old E_0 parameter. In the present closure it is no longer inserted as an external number. It is selected by Eq. (3) as an interior minimum of Eq. (2). The strongest statement justified by the batch data is therefore

$$\boxed{\text{the BEM/NLS finite-cell model derives a stable dimensionless aspect eigenvalue } E_\star}$$

from ideal-trefoil geometry plus the NLS-shell pressure closure.

However, the controls also show that the present evidence is not yet a proof of the electromagnetic fine-structure constant in QED. Two issues remain:

(i) The scalar BEM response is a proxy for the full Hodge–Steklov one-form problem.

(ii) The identification $\alpha_{\rm cell} = 2/E_{\rm eff}$ must be connected to a far-field interaction theorem.

The next required theorem is

$$V(r) = \frac{\alpha_{\rm cell}\hbar c}{r} + O(r^{-2}),$$

or equivalently

$$\frac{g_{\rm eff}^2}{4\pi\hbar c} = \alpha_{\rm cell}.$$

Without that theorem, the present result is best described as a dimensionless fine-structure-scale eigenvalue, not yet a complete derivation of QED electromagnetism.

8 Mechanism and stiffness asymptotics

The control runs show that the BEM term alone does not select a finite aspect eigenvalue. The finite stationary point is generated by the NLS pressure enthalpy, while the BEM contribution supplies a small compliance shift. This can be seen analytically from

$$\mathcal{A}_{\rm total}(E) = \mathcal{A}_{\rm BEM}(E) + \kappa\left[\frac{1}{3}\left(\frac{E}{E_p}\right)^3 - \ln\left(\frac{E}{E_p}\right)\right]. \tag{8}$$

Let $E = E_p + \Delta E$, with $|\Delta E| \ll E_p$. Since

$$\frac{d}{d\log E}\left[\frac{1}{3}\left(\frac{E}{E_p}\right)^3 - \ln\left(\frac{E}{E_p}\right)\right] = \left(\frac{E}{E_p}\right)^3 - 1 = 3\frac{\Delta E}{E_p} + O(\Delta E^2), \tag{9}$$

the stationarity condition gives

$$\boxed{\Delta E \simeq -\frac{E_p}{3\kappa}\,\frac{d\mathcal{A}_{\mathrm{BEM}}}{d\log E}\bigg|_{E_p}.} \tag{10}$$

Therefore

$$E_\star = E_p + O(\kappa^{-1}). \tag{11}$$

This is an important classification point: the numerical BEM calculation validates the finite-cell compliance shift around the NLS-shell pressure scale; it does not independently derive the prefactor $16\pi/3$, the shell coefficient $11/48$, or the pressure scale E_p. Those remain analytical closure ingredients supplied by the companion pressure-cell paper.

9 Far-field coupling theorem

The one-cell phase-Hessian normalization required to make this theorem independent is isolated in Appendix B. The appendix proves the exterior Green-function identity $\Lambda_\phi = 4\pi\mathcal{K}_{\mathrm{cell}}$ and states the remaining operator target $\Lambda_\phi = E_{\mathrm{eff}}/2$.

The result in this section is a conditional theorem. It proves the Coulombic $1/r$ form once the far-field stiffness normalization $\mathcal{K}_{\mathrm{cell}} = E_{\mathrm{eff}}/(8\pi)$ is supplied. The numerical two-cell certificate verifies the Green-function coefficient implied by that normalization. A fully independent microscopic derivation requires a one-cell phase-Hessian calculation that obtains $\Lambda_\phi = E_{\mathrm{eff}}/2$ without inserting the reduced quadratic action by hand.

The batch computation derives a dimensionless cell aspect value. To identify the result with the electromagnetic fine-structure constant, the cell must also produce a long-range interaction with the Coulomb Green-function tail. This section states the required far-field theorem in a form that is independent of electron mass, electric charge, the Rydberg constant, or CODATA input.

Let u denote the exterior massless boundary phase mode of the finite cell after the short-range core and pressure degrees of freedom have been integrated out. The far-field effective action is taken to be

$$\mathcal{A}_{\mathrm{far}}[u] = \hbar c\left[\frac{\mathcal{K}_{\mathrm{cell}}}{2}\int_{\mathbb{R}^3}|\nabla u|^2\,d^3x - \sum_{a=1}^{N} q_a u(\mathbf{x}_a)\right], \tag{12}$$

where $q_a \in \mathbb{Z}$ is the topological phase charge of cell a. The factor $\hbar c$ sets the conventional energy-length unit; it is not used to determine the dimensionless coupling. The dimensionless stiffness $\mathcal{K}_{\mathrm{cell}}$ is the only far-field normalization.

Lemma 1 (Poisson reduction)**.** *Stationarity of Eq. (12) gives*

$$-\mathcal{K}_{\mathrm{cell}}\nabla^2 u(\mathbf{x}) = \sum_a q_a\delta(\mathbf{x}-\mathbf{x}_a). \tag{13}$$

For two cells of unit charge at separation r, the interaction energy is

$$V(r) = \hbar c\,\frac{q_1 q_2}{4\pi\mathcal{K}_{\mathrm{cell}}}\frac{1}{r} + O(r^{-2}). \tag{14}$$

Proof. Varying Eq. (12) with respect to u, integrating by parts, and requiring the boundary term at infinity to vanish gives Eq. (13). In three dimensions,

$$-\nabla^2\left(\frac{1}{4\pi|\mathbf{x}-\mathbf{y}|}\right) = \delta(\mathbf{x}-\mathbf{y}). \tag{15}$$

Therefore the solution sourced by charge q_1 is

$$u_1(\mathbf{x}) = \frac{q_1}{4\pi\mathcal{K}_{\rm cell}|\mathbf{x}-\mathbf{x}_1|}. \tag{16}$$

Evaluating the source term of the second charge in this field gives Eq. (14), up to sign convention for like/opposite topological phase charges. Multipole and finite-size corrections begin at order r^{-2} or smaller depending on the imposed neutrality conditions. □

Theorem 1 (Far-field fine-structure certificate)**.** *If the exterior cell stiffness satisfies*

$$\mathcal{K}_{\rm cell} = \frac{E_{\rm eff}}{8\pi}, \tag{17}$$

then the unit-charge far-field interaction is

$$V(r) = \frac{\alpha_{\rm cell}\hbar c}{r} + O(r^{-2}), \qquad \alpha_{\rm cell} = \frac{2}{E_{\rm eff}}. \tag{18}$$

Equivalently,

$$\frac{g_{\rm eff}^2}{4\pi\hbar c} = \alpha_{\rm cell}. \tag{19}$$

Proof. Substitute Eq. (17) into Eq. (14) with $q_1q_2 = 1$. The coefficient is

$$\frac{1}{4\pi\mathcal{K}_{\rm cell}} = \frac{1}{4\pi(E_{\rm eff}/8\pi)} = \frac{2}{E_{\rm eff}} = \alpha_{\rm cell}. \tag{20}$$

Writing the interaction as $V(r) = g_{\rm eff}^2/(4\pi r)$, comparison with Eq. (18) gives Eq. (19). □

The theorem shows exactly what remains to be verified numerically or analytically: not the value of α itself, but the far-field stiffness normalization $\mathcal{K}_{\rm cell} = E_{\rm eff}/(8\pi)$. This can be tested without using m_e, e, R_∞, or CODATA.

10 Two-cell numerical certificate

A direct BEM certificate can be formulated by placing two identical cells at separation R, solving the finite-cell response for equal and opposite unit phase signs, and comparing the actions

$$\mathcal{A}_2^{++}(R), \qquad \mathcal{A}_2^{+-}(R). \tag{21}$$

The interaction coefficient is extracted from the charge-odd part,

$$C_{\rm far}(R) = \frac{R}{2}\left[\mathcal{A}_2^{++}(R) - \mathcal{A}_2^{+-}(R)\right]. \tag{22}$$

The far-field certificate is

$$\boxed{\lim_{R\to\infty} C_{\rm far}(R) = \frac{2}{E_{\rm eff}} = \alpha_{\rm cell}.} \tag{23}$$

A practical numerical table should therefore contain

$$R, \quad \mathcal{A}_2^{++}(R), \quad \mathcal{A}_2^{+-}(R), \quad C_{\rm far}(R), \quad \alpha_{\rm cell}, \quad \frac{C_{\rm far}(R)-\alpha_{\rm cell}}{\alpha_{\rm cell}}. \tag{24}$$

If $C_{\rm far}(R)$ converges to $\alpha_{\rm cell}$ under increasing separation and mesh refinement, then Eq. (18) is numerically certified. If the limit is absent or converges to a different value, then the eigenvalue calculation remains a dimensionless scale result but not a derivation of the electromagnetic coupling.

Remark 1 (Why ordinary closed-filament Biot–Savart tails are insufficient)**.** *A closed localized swirl-string by itself need not carry a monopolar $1/r$ field. Its ordinary velocity or Biot–Savart far field is generically multipolar. The Coulomb tail in Theorem 1 must therefore be carried by the massless exterior phase mode u, or equivalently by the Hodge–Steklov boundary phase sector, not by the short-range closed-filament velocity field alone.*

11 Derived-label status

The exact far-field stiffness gate is formulated in Appendix B; this appendix should be cited whenever the two-cell coefficient is described as conditional.

The present calculation justifies the label "closure-derived" for $E_\star$: given the NLS-shell pressure scale and stiffness, the BEM/NLS finite-cell action has a stable interior stationary point with controlled resolution and regularization dependence. It does not yet justify the stronger label "first-principles derived" for the electromagnetic fine-structure constant. That stronger label requires three additional analytical results:

1. a derivation of the pressure prefactor $16\pi/3$;
2. a derivation of the finite-shell coefficient $11/48$;
3. an independent one-cell phase-Hessian derivation of $\mathcal{K}_{\text{cell}} = E_{\text{eff}}/(8\pi)$.

The accompanying audit script `audit_derived_label_gates.py` encodes these requirements as explicit pass/fail gates.

12 Conclusion

The batch computation supports the following chain:

$$K = 3_1 \longrightarrow \mathcal{L}_K \longrightarrow \left(E_p^{\text{NLS}}, \kappa_{\text{NLS}}\right) \longrightarrow \delta_E \mathcal{A}_{\text{total}} = 0 \longrightarrow E_\star \longrightarrow \alpha_{\text{cell}}.$$

No electron mass, electric charge, Planck constant, Rydberg constant, or CODATA value is used in the derivation of $E_\star$. Across 16 runs, the model yields $E_\star = 274.074903758 \pm 5.51e-05$ and $\alpha_{\text{cell}}^{-1} = 137.035953090 \pm 2.76e-05$. This is strong evidence that the missing aspect scale can be obtained from a finite-cell eigenvalue problem. A final claim that α_{cell} is the electromagnetic fine-structure constant requires the far-field coupling certificate.

A Full batch table

B Far-field stiffness derivation target

This appendix isolates the remaining far-field normalization gate. The two-cell calculation verifies the Green-function coefficient once the stiffness $\mathcal{K}_{\text{cell}}$ is supplied. To upgrade the far-field result from an internal certificate to an independent derivation, $\mathcal{K}_{\text{cell}}$ must be obtained from a one-cell phase-Hessian operator.

B.1 Lemma B1: one-cell phase Hessian

Let u be the exterior massless phase field of a normalized cell. In units of $\hbar c$, write

$$\mathcal{A}_{\text{far}}[u] = \frac{\mathcal{K}_{\text{cell}}}{2} \int_{\mathbb{R}^3 \setminus B_1} |\nabla u|^2 \, d^3x. \tag{25}$$

For the exterior monopole profile

$$u(r) = \frac{\phi}{r}, \qquad r \geq 1, \tag{26}$$

one has

$$\int_{r \geq 1} |\nabla u|^2 \, d^3x = 4\pi\phi^2. \tag{27}$$

Therefore

$$\mathcal{A}_{\rm far}(\phi) = 2\pi\mathcal{K}_{\rm cell}\phi^2. \tag{28}$$

Define the one-cell phase Hessian by

$$\mathcal{A}_{\rm cell}(\phi) = \mathcal{A}_{\rm cell}(0) + \frac{1}{2}\Lambda_\phi\phi^2 + O(\phi^4). \tag{29}$$

Matching Eqs. (28) and (29) gives

$$\Lambda_\phi = 4\pi\mathcal{K}_{\rm cell}. \tag{30}$$

Hence

$$\mathcal{K}_{\rm cell} = \frac{E_{\rm eff}}{8\pi} \quad\Longleftrightarrow\quad \Lambda_\phi = \frac{E_{\rm eff}}{2}. \tag{31}$$

Proof. Since

$$|\nabla(\phi/r)|^2 = \frac{\phi^2}{r^4},$$

one obtains

$$\int_1^\infty 4\pi r^2\frac{\phi^2}{r^4}\,dr = 4\pi\phi^2.$$

This proves Eq. (27). The remaining relations follow by comparing the quadratic coefficient in ϕ. □

Derived-label gate. The identity $\Lambda_\phi = E_{\rm eff}/2$ must be produced by a one-cell phase-Hessian operator. If instead one imposes the reduced action $\mathcal{A}_\phi = E_{\rm eff}\phi^2/4$, then Eq. (31) is an internal normalization certificate, not an independent microscopic derivation. The required future calculation is therefore

$$\Lambda_\phi = \left.\frac{\partial^2\mathcal{A}_{\rm cell}[\phi]}{\partial\phi^2}\right|_{\phi=0} \overset{?}{=} \frac{E_{\rm eff}}{2},$$

with the left-hand side obtained from the cell operator rather than inserted as a normalization.

B.2 Connection to the two-cell coefficient

If Eq. (31) is independently established, then the two-cell Green-function interaction gives

$$V(r) = \hbar c\,\frac{q_1q_2}{4\pi\mathcal{K}_{\rm cell}}\frac{1}{r} + O(r^{-2}). \tag{32}$$

For unit topological charges,

$$\frac{1}{4\pi\mathcal{K}_{\rm cell}} = \frac{2}{E_{\rm eff}} = \alpha_{\rm cell}, \tag{33}$$

and therefore

$$V(r) = \frac{\alpha_{\rm cell}\hbar c}{r} + O(r^{-2}). \tag{34}$$

This is the precise condition under which the far-field part of Paper III may be upgraded from conditional theorem to independent coupling derivation.

References

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Table 3: Main convergence table exported from `batch_desired_table.csv`.

Grid $n_s : n_\theta : N_{\text{sph}}$	ϵ/a	$E_\star$	$E_\star - E_p$	$\alpha^{-1}_{\text{cell}}$	Rel. error
100:14:300	0.04	274.074821720	-5.39e-05	137.035912070	6.36e-07
100:14:300	0.05	274.074842923	-3.27e-05	137.035922672	5.58e-07
100:14:300	0.06	274.074865807	-9.85e-06	137.035934114	4.75e-07
100:14:300	0.08	274.074914752	3.91e-05	137.035958587	2.96e-07
120:16:360	0.04	274.074841166	-3.45e-05	137.035921793	5.65e-07
120:16:360	0.05	274.074868090	-7.57e-06	137.035935255	4.66e-07
120:16:360	0.06	274.074896280	2.06e-05	137.035949351	3.64e-07
120:16:360	0.08	274.074954626	7.90e-05	137.035978524	1.51e-07
140:18:420	0.04	274.074861365	-1.43e-05	137.035931893	4.91e-07
140:18:420	0.05	274.074893423	1.78e-05	137.035947922	3.74e-07
140:18:420	0.06	274.074926205	5.05e-05	137.035964313	2.54e-07
140:18:420	0.08	274.074992336	1.17e-04	137.035997379	1.31e-08
160:20:480	0.04	274.074881494	5.84e-06	137.035941957	4.18e-07
160:20:480	0.05	274.074918027	4.24e-05	137.035960224	2.84e-07
160:20:480	0.06	274.074954705	7.90e-05	137.035978563	1.50e-07
160:20:480	0.08	274.075027208	1.52e-04	137.036014815	-1.14e-07